

DevOps Linux Fundamentals (Day 7) - Machine Test

Real-World Scenario: You have joined as a Junior DevOps Engineer. A production Ubuntu server requires package management, storage verification, mount management, backup operations, and troubleshooting. Complete all tasks and submit command outputs/screenshots where required.

Part 1 - Package Management

1. Update package repository information.

sudo apt update

2. Install tree, curl, and git packages.

```
sreelekha@LAPTOP-CKI0PC9R:~$ sudo apt install tree curl git
tree is already the newest version (2.3.1-1).
curl is already the newest version (8.18.0-1ubuntu2.7).
curl set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 37
```

3. Search for Docker-related packages.

apt search docker

```
sreelekha@LAPTOP-CKI0PC9R:~$ apt search docker
amazon-ecr-credential-helper/resolute 0.7.1-2.1build1 amd64
  Amazon ECR Credential Helper for Docker

auto-apt-proxy/resolute 17.1 all
  automatic detector of common APT proxy settings

buildah/resolute 1.42.1+ds1-2build1 amd64
  CLI tool to facilitate building OCI images

catatonit/resolute 0.2.1-2build1 amd64
  init process for containers

centreon-plugins/resolute 0.0-20250303+dfsg-1 all
  Collection of Nagios plugins to monitor OS, services and network

crowdsec/resolute 1.4.6-10.1 amd64
  lightweight and collaborative security engine

crun/resolute 1.21-1ubuntu3 amd64
  lightweight OCI runtime for running containers

cwl-utils/resolute 0.40-1 all
  Utilities for using CWL documents

debcraft/resolute 0.8.1 all
  Easy, fast and secure way to build Debian packages

debocker/resolute 0.2.6build1 all
  docker-powered package builder for Debian
lines 1-25...skipping...
amazon-ecr-credential-helper/resolute 0.7.1-2.1build1 amd64
```

4. Display package information for nginx.

apt show nginx

```
sreelekha@LAPTOP-CKI0PC9R:~$ apt show nginx
Package: nginx
Version: 1.28.3-2ubuntu1.11
Priority: optional
Section: web
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Debian Nginx Maintainers <pkg-nginx-maintainers@alioth-lists.debian.net>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 1631 kB
Provides: httpd, httpd-cgi, nginx-abi-1.28.3-1
Depends: libc6 (>= 2.34), libcrypt1 (>= 1:4.1.0), libpcre2-8-0 (>= 10.22), libssl3t64 (>= 3.0.0), zlib1g (>= 1:1.1.4), nginx-common (= 1.28.3-2ubuntu1.11)
Breaks: nginx-core (<< 1.22.1-6~), nginx-extras (<< 1.22.1-6~), nginx-light (<< 1.22.1-6~)
Replaces: nginx-core (<< 1.22.1-6~), nginx-extras (<< 1.22.1-6~), nginx-light (<< 1.22.1-6~)
Homepage: https://nginx.org
Download-Size: 617 kB
APT-Manual-Installed: yes
APT-Sources: http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
Description: small, powerful, scalable web/proxy server
 Nginx ("engine X") is a high-performance web and reverse proxy server
 created by Igor Sysoev. It can be used both as a standalone web server
 and as a proxy to reduce the load on back-end HTTP or mail servers.

Notice: There is 1 additional record. Please use the '-a' switch to see it
```

5. Remove tree package.

`sudo apt remove tree`

```
sreelekha@LAPTOP-CKI0PC9R:~$ sudo apt remove tree
REMOVING:
  tree

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 37
  Freed space: 125 kB

Continue? [Y/n] y
(Reading database ... 45591 files and directories currently installed.)
Removing tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
```

Evidence: Capture outputs of apt search docker and apt show nginx.

Part 2 - Storage Analysis

1. Check filesystem usage using `df -h`.

```
sreelekha@LAPTOP-CKI0PC9R: ~
sreelekha@LAPTOP-CKI0PC9R:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
none	1.9G	0	1.9G	0%	/usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none	1.9G	4.0K	1.9G	1%	/mnt/wsl
drivers	205G	165G	40G	81%	/usr/lib/wsl/drivers
/dev/sdd	1007G	2.4G	954G	1%	/
none	1.9G	36K	1.9G	1%	/mnt/wslg
none	1.9G	0	1.9G	0%	/usr/lib/wsl/lib
rootfs	1.9G	2.8M	1.9G	1%	/init
none	1.9G	556K	1.9G	1%	/run
none	1.9G	0	1.9G	0%	/run/lock
none	1.9G	0	1.9G	0%	/run/shm
none	1.9G	80K	1.9G	1%	/mnt/wslg/versions.txt
none	1.9G	80K	1.9G	1%	/mnt/wslg/doc
C:\	205G	165G	40G	81%	/mnt/c
D:\	138G	3.2G	135G	3%	/mnt/d
E:\	135G	2.2G	133G	2%	/mnt/e
tmpfs	1.9G	0	1.9G	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	374M	12K	374M	1%	/run/user/1000

2. Check project folder size using `du -sh project`.

```
sreelekha@LAPTOP-CKI0PC9R: ~
sreelekha@LAPTOP-CKI0PC9R:~$ du -sh project
8.0K    project
sreelekha@LAPTOP-CKI0PC9R:~$
```

3. View storage devices using lsblk.

```
sreelekha@LAPTOP-CKI0PC9R:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    1G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /var/lib/docker
                               /mnt/wslg/distro
                               /

sreelekha@LAPTOP-CKI0PC9R:~$
```

4. Display UUID information using sudo blkid.

```
sreelekha@LAPTOP-CKI0PC9R:~$ sudo blkid
[sudo: authenticate] Password:
/dev/sdd: UUID="7ac6f8f4-f5b5-4195-8977-5f27d0815a66" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdb: BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdc: UUID="f50c65e2-9700-4344-b4ec-1f6397312828" TYPE="swap"
/dev/sda: BLOCK_SIZE="4096" TYPE="ext4"
sreelekha@LAPTOP-CKI0PC9R:~$
```

5. Display partition details using sudo fdisk -l.

```
sreelekha@LAPTOP-CKI0PC9R:~$ sudo fdisk -l
Disk /dev/ram0: 4 MiB, 4194304 bytes, 8192 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/ram1: 4 MiB, 4194304 bytes, 8192 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/ram2: 4 MiB, 4194304 bytes, 8192 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/ram3: 4 MiB, 4194304 bytes, 8192 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/ram4: 4 MiB, 4194304 bytes, 8192 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
```

Save outputs into storage-report.txt.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ df -h > storage-report.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ du -sh project >> storage-report.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ lsblk >> storage-report.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ sudo blkid >> storage-report.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ sudo fdisk -l >> storage-report.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ cat storage-report.txt  
Filesystem      Size  Used Avail Use% Mounted on  
none            1.9G  0  1.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2  
none            1.9G  4.0K  1.9G   1% /mnt/wsl  
drivers          205G  164G   41G  81% /usr/lib/wsl/drivers  
/dev/sdd         1007G  2.4G  954G   1% /  
none            1.9G  48K  1.9G   1% /mnt/wslg  
none            1.9G  0  1.9G   0% /usr/lib/wsl/lib  
rootfs          1.9G  2.8M  1.9G   1% /init  
none            1.9G  560K  1.9G   1% /run  
none            1.9G  0  1.9G   0% /run/lock  
none            1.9G  0  1.9G   0% /run/shm  
none            1.9G  80K  1.9G   1% /mnt/wslg/versions.txt  
none            1.9G  80K  1.9G   1% /mnt/wslg/doc  
C:\              205G  164G   41G  81% /mnt/c  
D:\              138G  3.2G  135G   3% /mnt/d  
E:\              135G  2.2G  133G   2% /mnt/e  
tmpfs            1.9G  0  1.9G   0% /tmp  
none            1.0M  0  1.0M   0% /run/credentials/systemd-journald.service  
none            1.0M  0  1.0M   0% /run/credentials/getty@tty1.service  
tmpfs            374M  12K  374M   1% /run/user/1000  
none            1.0M  0  1.0M   0% /run/credentials/systemd-resolved.service  
8.0K            project  
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda    8:0    0 356.9M  1 disk  
sdb    8:16   0 159.4M  1 disk  
sdc    8:32   0    1G  0 disk [SWAP]  
sdd    8:48   0    1T  0 disk /var/lib/docker  
                               /mnt/wslg/distro  
                               /  
/dev/sdd: UUID="7ac6f8f4-f5b5-4195-8977-5f27d0815a66" BLOCK_SIZE="4096" TYPE="ext4"  
/dev/sdb: BLOCK_SIZE="4096" TYPE="ext4"  
/dev/sdc: UUID="f50c65e2-9700-4344-b4ec-1f6397312828" TYPE="swap"  
/dev/sda: BLOCK_SIZE="4096" TYPE="ext4"
```

Part 3 - Mounting & Filesystem Management

1. Create a mount point: `sudo mkdir /mnt/devops-data`
2. Display current mount information using `mount`.

```
sreelekha@LAPTOP-CKI0PC9R:~$ sudo mkdir /mnt/devops-data  
sreelekha@LAPTOP-CKI0PC9R:~$ mount  
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uid=on)  
none on /mnt/wsl type tmpfs (rw,relatime)  
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)  
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)  
none on /mnt/wslg type tmpfs (rw,relatime)  
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)  
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uid=on)  
rootfs on /init type rootfs (ro,size=1987904k,nr_inodes=476976)  
none on /dev type devtmpfs (rw,nosuid,relatime,sizes=1987904k,nr_inodes=476976,mode=755)  
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)  
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)  
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)  
none on /run type tmpfs (rw,nosuid,nodev,mode=755)  
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)  
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)  
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)  
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)  
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)  
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)  
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/system/vhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)  
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/system/vhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)  
none on /tmp/.X11-unix type tmpfs (ro,relatime)  
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)  
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)  
E:\ on /mnt/e type 9p (rw,noatime,aname=drvfs;path=E:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)  
none on /run/user type tmpfs (rw,relatime)  
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)  
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)  
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)  
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)  
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,nr_inodes=1048576)  
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)  
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)  
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)  
none on /tmp/.X11-unix type tmpfs (ro,relatime)  
/dev/sdd on /var/lib/docker type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)  
none on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
```

3. Check mounted filesystems using df -h.

```
sreelekha@LAPTOP-CKI0PC9R:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G   0    1.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            1.9G  4.0K   1.9G   1% /mnt/wsl
drivers         205G  164G   42G   80% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.4G  954G   1% /
none            1.9G   48K   1.9G   1% /mnt/wslg
none            1.9G   0    1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  2.8M   1.9G   1% /init
none            1.9G  560K   1.9G   1% /run
none            1.9G   0    1.9G   0% /run/lock
none            1.9G   0    1.9G   0% /run/shm
none            1.9G   80K   1.9G   1% /mnt/wslg/versions.txt
none            1.9G   80K   1.9G   1% /mnt/wslg/doc
C:\             205G  164G   42G   80% /mnt/c
D:\             138G  3.2G  135G   3% /mnt/d
E:\             135G  2.2G  133G   2% /mnt/e
tmpfs            1.9G   0    1.9G   0% /tmp
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs            374M  12K   374M   1% /run/user/1000
none            1.0M   0    1.0M   0% /run/credentials/systemd-resolved.service
```

4. View automatic mount configuration using cat /etc/fstab.

```
sreelekha@LAPTOP-CKI0PC9R:~$ cat /etc/fstab
# UNCONFIGURED FSTAB FOR BASE SYSTEM
```

5. Create mount-report.txt containing mount, df -h, and /etc/fstab output.

```
sreelakha@LAPTOP-CKI0PC9R:~$ touch mount-report.txt
sreelakha@LAPTOP-CKI0PC9R:~$ mount > mount-report.txt
sreelakha@LAPTOP-CKI0PC9R:~$ df -h >> mount-report.txt
sreelakha@LAPTOP-CKI0PC9R:~$ cat /etc/fstab >> mount-report.txt
sreelakha@LAPTOP-CKI0PC9R:~$ cat mount-report.txt
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers,fmask=222,dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uid=on)
rootfs on /init type rootfs (ro,size=1907904k,nr_inodes=476976)
none on /dev type devtmpfs (rw,nosuid,relatime,size=1907904k,nr_inodes=476976,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
E:\ on /mnt/e type 9p (rw,noatime,aname=drvfs;path=E:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,nr_inodes=1048576)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
/dev/sdd on /var/lib/docker type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=382732k,nr_inodes=95683,mode=700,uid=1000,gid=1000)
tmpfs on /mnt/wslg/run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=382732k,nr_inodes=95683,mode=700,uid=1000,gid=1000)
none on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G  0 1.9G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            1.9G  4.0K 1.9G  1% /mnt/wsl
drivers         205G  164G  42G  80% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.4G  954G  1% /
none            1.9G  48K 1.9G  1% /mnt/wslg
none            1.9G  0 1.9G  0% /usr/lib/wsl/lib
rootfs          1.9G  2.0M 1.9G  1% /init
none            1.9G  560K 1.9G  1% /run
none            1.9G  0 1.9G  0% /run/lock
none            1.9G  0 1.9G  0% /run/shm
none            1.9G  80K 1.9G  1% /mnt/wslg/versions.txt
none            1.9G  80K 1.9G  1% /mnt/wslg/doc
C:\             205G  164G  42G  80% /mnt/c
D:\             138G  3.2G 135G  3% /mnt/d
E:\             135G  2.2G 133G  2% /mnt/e
tmpfs           1.9G  0 1.9G  0% /tmp
none            1.0M  0 1.0M  0% /run/credentials/systemd-journald.service
none            1.0M  0 1.0M  0% /run/credentials/getty@tty1.service
tmpfs           374M  12K 374M  1% /run/user/1000
none            1.0M  0 1.0M  0% /run/credentials/systemd-resolved.service
# UNCONFIGURED FSTAB FOR BASE SYSTEM
sreelakha@LAPTOP-CKI0PC9R:~$
```

Part 4 - Backup & Compression

1. Create TAR backup.

```
tar -cvf project-backup.tar project/
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ tar -cvf project-backup.tar project/project/  
project/.env  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -lh project-backup.tar  
-rw-r--r-- 1 sreelekha sreelekha 10K Oct  7 09:16 project-backup.tar
```

2. Extract TAR backup.

```
tar -xvf project-backup.tar
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ tar -xvf project-backup.tar  
project/  
project/.env  
sreelekha@LAPTOP-CKI0PC9R:~$
```

3. Create compressed archive.

```
tar -czvf project-backup.tar.gz project/
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ tar -czvf project-backup.tar.gz project/  
project/  
project/.env  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -lh project-backup.tar.gz  
-rw-r--r-- 1 sreelekha sreelekha 161 Oct  7 09:30 project-backup.tar.gz  
sreelekha@LAPTOP-CKI0PC9R:~$
```

4. Extract compressed archive.

```
tar -xzvf project-backup.tar.gz
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ tar -xzvf project-backup.tar.gz  
project/  
project/.env  
sreelekha@LAPTOP-CKI0PC9R:~$
```

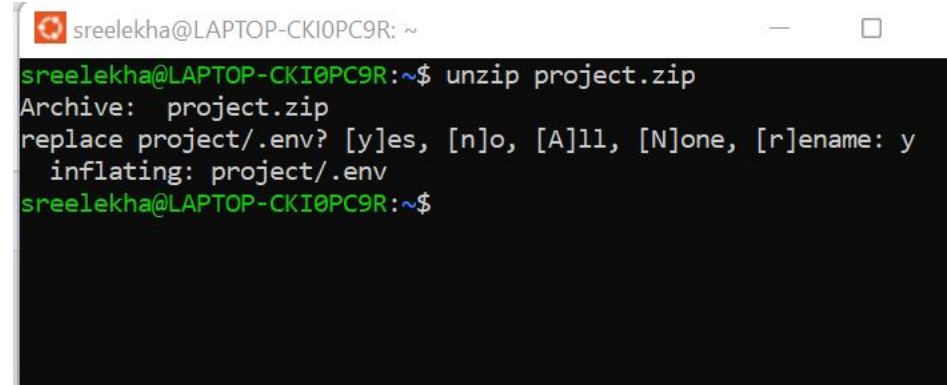
5. Create ZIP backup.

```
zip -r project.zip project/
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ zip -r project.zip project/  
adding: project/ (stored 0%)  
adding: project/.env (deflated 50%)  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -lh project.zip  
-rw-r--r-- 1 sreelekha sreelekha 324 Oct  7 09:33 project.zip  
sreelekha@LAPTOP-CKI0PC9R:~$
```

6. Extract ZIP backup.

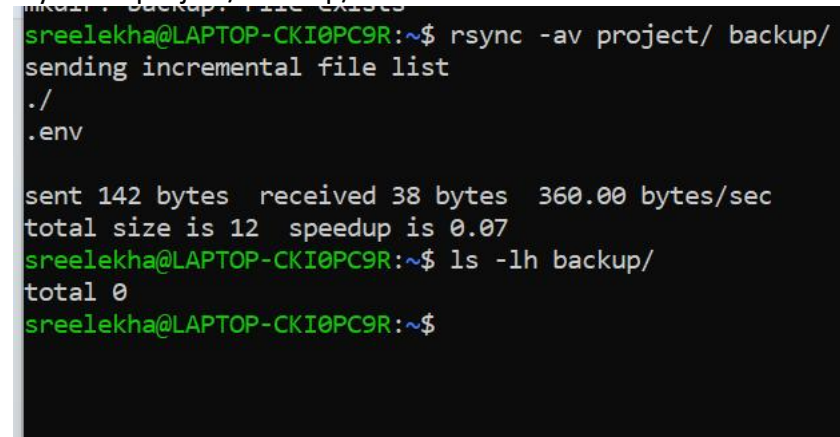
unzip project.zip



```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ unzip project.zip  
Archive:  project.zip  
replace project/.env? [y]es, [n]o, [A]ll, [N]one, [r]ename: y  
inflating: project/.env  
sreelekha@LAPTOP-CKI0PC9R:~$
```

7. Backup application using rsync.

rsync -av project/ backup/



```
sreelekha@LAPTOP-CKI0PC9R:~$ rsync -av project/ backup/  
sending incremental file list  
./  
.env  
  
sent 142 bytes  received 38 bytes  360.00 bytes/sec  
total size is 12  speedup is 0.07  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -lh backup/  
total 0  
sreelekha@LAPTOP-CKI0PC9R:~$
```

Part 5 - Production Troubleshooting Challenge

Scenario:

Filesystem Size Used Avail Use%
/dev/sda1 50G 49G 1G 98%

Answer:

1. Which command identifies disk usage?

Ans) df -h

2. Which command shows folder size?

Ans) du -sh folder_name

3. Which command shows available disks?

Ans) lsblk

4. Which command removes unused packages?

Ans) sudo apt autoremove

5.Which command is commonly used for production backups?

Ans)rsync -av source/ backup/

Bonus Challenge

Create a backup workflow:

mkdir deployment-backup

rsync -av project/ deployment-backup/

tar -czvf deployment-backup.tar.gz deployment-backup/

Verify using:

ls -lh deployment-backup.tar.gz

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ mkdir deployment-backup  
mkdir: deployment-backup: File exists  
sreelekha@LAPTOP-CKI0PC9R:~$ rsync -av project/ deployment-backup/  
sending incremental file list  
  
sent 80 bytes  received 12 bytes  184.00 bytes/sec  
total size is 12  speedup is 0.13  
sreelekha@LAPTOP-CKI0PC9R:~$ tar -czvf deployment-backup.tar.gz deployment-backup/  
deployment-backup/  
deployment-backup/.env  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -lh deployment-backup.tar.gz  
-rw-r--r-- 1 sreelekha sreelekha 176 Oct  7 09:56 deployment-backup.tar.gz  
sreelekha@LAPTOP-CKI0PC9R:~$
```

Interview Questions

1. What is APT?

APT (Advanced Package Tool) is a package management tool used in Debian-based Linux systems like Ubuntu to install, update, remove, and manage software packages.It is used to **install, update, upgrade, and remove software packages.**

```
sudo apt install nginx
```

2. Difference between apt update and apt upgrade?

- apt update → Updates the package repository information.
- apt upgrade → Installs the latest available versions of installed packages.

Simple way to remember:

apt update = Updates package information

apt upgrade = Updates installed packages

3. What does df -h display?

df -h displays **filesystem disk space usage** in a human-readable format.

It shows:

- Total space
 - Used space
 - Available space
 - Usage percentage
 - Mount point
-

4. Difference between df -h and lsblk?

df -h shows **disk space usage** of mounted filesystems.

lsblk shows **storage devices and partitions** available on the system.

5. What is UUID?

UUID (Universally Unique Identifier) is a unique identifier assigned to a filesystem or partition.

It is commonly used in /etc/fstab to identify a specific partition for automatic mounting.

Example:

UUID="1234-abcd"

6. What is mounting in Linux?

Mounting means making a storage device or filesystem accessible through a directory in Linux.

Example:

```
sudo mount /dev/sdb1 /mnt/data
```

Here:

- /dev/sdb1 → Storage device/partition
- /mnt/data → Mount point

7. Difference between mount and umount?

- mount → Attaches a filesystem to the Linux directory structure.
- umount → Detaches a mounted filesystem.

Example:

```
sudo mount /dev/sdb1 /mnt/data  
sudo umount /mnt/data
```

8. What is /etc/fstab?

/etc/fstab is a Linux configuration file that contains information about **filesystems that should be mounted automatically**, usually during system boot.

It can contain:

- UUID
 - Mount point
 - Filesystem type
 - Mount options
-

9. Difference between .tar and .tar.gz?

.tar

- Combines multiple files and directories into one archive.
- Does **not** compress the data.

.tar.gz

- Creates a TAR archive and then compresses it using **gzip**.
- Usually has a smaller file size.

Easy way:

```
.tar = Archive  
.tar.gz = Archive + Compression
```

10. Why is rsync preferred in production?

rsync is commonly preferred for production backups because it:

- Transfers only changed files.
- Saves time and bandwidth.
- Supports incremental backups.
- Preserves file permissions and timestamps.
- Can synchronize files between local and remote systems.
- Works efficiently with large amounts of data.

Example:

```
rsync -av project/ backup/
```